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TECHNICAL NOTE

Criminalistics

Relevant propositions for Y chromosome interpretation

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Abstract

The Y chromosomal haplotype is expected to be identical (or close to, depending on the mutation rate) among a male and many of his paternal relatives. This means that often the same evidential value for the DNA evidence is obtained, whether the true donor or one of his close paternal relatives is compared to a crime sample. Commentators (see for example the UK Forensic Science Regulator or Amorim) have suggested to change the proposition pair to compare the probability of the evidence if the Person of Interest (POI) or one of his close paternal relatives left the DNA to the probability of the evidence if an unrelated male from the population left the DNA. We argue that this is problematic because there is no clear definition of close paternal relatives and truly unrelated males do not exist. Instead, we take a starting point in the traditional proposition pair "The source of the male DNA is the POI" versus "The source of the male DNA is not the POI" and make the latter one operational by suggesting that it is formulated as "The source of the male DNA is a random man from the population". The issue of matching males in the POI's lineage is then addressed either in a comment in the statement or directly through a probability model.

KEYWORDS
casework, forensic Y chromosome analysis, interpretation, likelihood ratios, propositions, relatedness

Highlights

- Relevant proposition pairs for Y-STR forensic DNA evidence for assigning an *LR* are discussed.
- This proposition set aligns with the typical calculations undertaken for *LR*s for Y-STR evidence.
- We highlight the challenges of defining paternal relatives.

1 | INTRODUCTION

In the forensic interpretation of single source Y chromosome haplotypes, a sample associated with a crime, G_C, is compared with a sample from a person of interest (POI) G_P [1, 2]. If these two samples have the same alleles at each locus, then the crime sample could come from the POI. There, are however, other persons from whom the crime sample could have originated and those are all people with the same haplotype (which a priori could be anyone in the pool of potential alternate donors).

All Y chromosomes are descended from one ancestral chromosome. In that sense all modern Y chromosomes are related [3]. Many differ now because of mutations that have occurred in the intervening generations. The probability of a mutation (μ) at one or more of the 23 loci in, say, the forensic multiplex Promega PowerPlex® Y23 is about 0.074 per generation [4] (i.e., approximately one out of every 14 father-son pairs will possess at least one mutation). It follows that close patrilineal relatives may or may not have the same haplotype while quite distant male relatives may still have the same haplotype [5]. If the genotyping has been undertaken with a modern

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multiplex that types 23 or more loci, most probably all people with the same haplotype will be related to the POI within some number of male-line links that can be a few to over 10 [5]. This has been shown by simulation. See for example Andersen and Balding [5] who state "Our results confirm that in practice matches only occur between males connected by a lineage path of at most a few tens of meioses."

Consider a group of people having the same Y-haplotype at, say, 23 loci. Call this the central haplotype. The probability that one of these males mutates away from this haplotype at each meiosis is about 7.4%. We can imagine that, over time, this group spreads out into a clade of similar haplotypes differing at one or a few loci. There is a chance of back mutation to the central haplotype, but this requires, e.g. for example, that the difference is at one locus and that this exact locus mutates directly back to the central haplotype. Next, consider an entirely different lineage. There is only a small chance that a haplotype from a male from this lineage is even remotely similar to the central haplotype described above.

When the haplotype developed from the crime scene sample is the same as the POI, it is usual to assign a weight to the evidence. This is often presented in the form of a likelihood ratio, *LR*, but as suggested and illustrated by [5], this can also be in terms of reporting the number of matching males. When assigning an *LR*, (at least) two mutually exclusive propositions are required. With single source profiles, many laboratories report a match probability and not an *LR*. Although this also uses a proposition, that is seldom explicitly stated.

In forensic genetics, we are often interested in the following two mutually exclusive and exhaustive propositions:

> **Proposition 1.** The source of the male DNA is the POI.

> **Proposition 2.** The source of the male DNA is not the POI.

Often, it is possible to evaluate the probability of the evidence under Proposition 1, but Proposition 2 is in a generic state that does not enable computations. Instead, this needs to be made concrete such that computations can be performed. One such example is that the source of the male DNA is a random man from a relevant reference population. This brings us to the topic of this paper.

We are aware that the propositions used by some practitioners to assign the *LR* for Y chromosome evidence are of the form:

> **Proposition 1.** The source of the male DNA is [Suspect name] (or a close paternal-line male relative of his).

> **Proposition 2.** The source of the male DNA is an unrelated male from the [reference population].

The specific two propositions given above come from the UK Forensic Science Regulator Guidance for Y-STR profiling [1] (hereafter UK Regulator) but we have seen them elsewhere in court transcripts and reports which we do not explicitly reference here as to focus the discussion on academic discussion and institutional policies rather than personal testimony.

The UK Regulator also gives advice on phrasing when reporting a verbal qualifier in conjunction with an *LR*. They suggest to report: "In my opinion, the scientific findings provide [degree of support] for the view that [Suspect name] (or a close paternal-line male relative of his) deposited the male DNA on the [crime stain item] rather than an unknown unrelated man."

Amorim proposes a similar proposition pair [2]: "The situation is completely different if lineage markers are considered. First of all, the very formulation of the hypotheses should be changed into

> Hp. *The suspect (or someone of the same lineage) is the donor of crime scene sample*.

> Hd. *The suspect is not the donor of crime scene sample or anybody from the same lineage"*.

In this commentary, we present an argument for why these propositions are not suitable and offer an argument for a more relevant proposition pair. We also review the literature for statements about the propositions to use.

2 | LITERATURE REVIEW

We searched the literature for papers and guidance documents that had previously considered propositions for Y chromosome evidence at the sub-source level within the hierarchy of propositions. Other than the UK Regulator and Amorim we found very few articles that clearly state the propositions. This is in contrast to autosomal evidence where a number of papers and guidance documents exist specifying proposition at the sub-source level [6–8].

Some of the articles we found did state the propositions without comment. These include the following:

Andersen et al. [9] state that in cases without close male relatives to the suspect (with a Y-STR profile matching the DNA evidence from, e.g., a crime scene), the following propositions can be used:

> Hs. *The hypothesis that the DNA is from a suspect, and*

> Hr. *The hypothesis that the DNA from the crime scene is from a random man*.

The DNA Commission of the International Society of Forensic Genetics (ISFG) 2020 [10] state:

> **Hypothesis H1.** *Person X is the source of the DNA*.

> **Hypothesis H2.** *A random man Y from the reference population Z is the source of the DNA, and Y is another man than X*.

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The DNA Commission paper additionally gives a suggested wording that we reproduce below:

The Y-STR profile detected in the crime stain is LR times more probable to observe under hypothesis H1 than under hypothesis H2. This notwithstanding, paternal relatives have a high probability to have the same Y-STR profile and will in that case have the same likelihood ratio (LR).

Andersen and Balding state [11]: Hypotheses of interest are as follows:

HQ. *The evidence profile came from Q*,

HX. *The evidence profile came from X*,

where X is an alternative to Q as the source of the DNA whose profile is not available.

The remaining articles that we found gave, to our reading, a cryptic definition that is consistent with the set of propositions that we give later and suggest are relevant.

Caliebe and Krawczak [12] "address one of the key requirements of LR calculations, namely to account properly for the possible relatedness of suspect and trace donor, if H_d [the standpoint of the defence] is correct."

The DNA Commission of the ISFG 2001 [13] state "The usual assumption is that the suspect, if innocent, has been selected from the population in a way that gives him only a random chance to be related to the perpetrator."

Wolf et al. [14] describe "the likelihood ratio $L(E)$ of donorship of the suspect versus non-donorship of the suspect". Similarly, Krawczak [15] states "inference making in forensics has generally been accepted to be based upon the likelihood ratio $L(E)$ of the prosecutor and defence hypothesis, given the evidence E."

3 | DISCUSSION

We return to the propositions detailed by the UK Regulator:

H1. *The source of the male DNA is [Suspect name] (or a close paternal-line male relative of his).*

H2. *The source of the male DNA is an unrelated male from the [reference population].*

We have three objections to these propositions. These are as follows:

1. These propositions are not what is under consideration—that is, they do not reflect the position of the prosecution (that [Person of Interest] is the donor of the male DNA) and therefore are not addressing the question required by the court or fact-finder.
2. There is no agreed, viable definition of close patrilineal relative and truly unrelated males do not exist.
3. They cannot be implemented.

We discuss these in reverse order.

Consider an attempt to assess the *LR* where H_1 and H_2 are propositions 1 and 2 as given by the UK Regulator, respectively, and I is all relevant background information. This is usually given as the inverse of the match probability where the match probability is based on a count, modified count, or some other function of data in a population sample. However, we point out that the numerator of the *LR* is not strictly 1 since no close patrilineal relatives are certain to have the same haplotype due to the probability of one or more intervening mutations. The alternative donor in H_2 is defined as an unrelated male. To estimate the probability of obtaining the observed profile for an unrelated male from the reference population, we would need to remove from the population sample all relatives of the POI. The probable outcome, if it was possible, of removing all relatives from the database, skipping over the problem of defining relatives, is that the sample count of this haplotype is now zero, as very distantly related males are unlikely to share a haplotype.

The UK Regulator guidance does not explicitly state how to assign the numerator or denominator of the *LR* using their suggested proposition set. However, the general content does appear to suggest using $LR = \frac{1}{\Pr(G_S|H_2,\ I)}$ where $\Pr(G_S|H_2,I)$ is "recommended to use the $(n+1)/(N+1)$ approach until alternative methods emerge that have been demonstrated (through validation) to be a robust suitable replacement." where n is the count of the haplotype in a database of size N. This would suggest that they are implementing the correct set of propositions, despite their recommended propositions.

In terms of our second objection, with regard to the impossibility of defining close male relatives, we note the obvious that every meiosis separating two individuals who have a common ancestor increases the probability that their haplotypes are different.

Andersen and Balding state that for modern Y-profiling kits then "Unless μ is very low, the possibility of a match between distantly-related individuals is negligible relative to the much more likely event of a match between pairs of individuals who are related closely enough to generate the match, but distantly enough that the relatedness is not recognised." [11].

The last point is that these are not the propositions of interest to the court. The court is interested in whether or not the POI is the donor to the stain. If the Bayesian approach is applied by the court, which of course the court is under no obligation to do, then prior odds are needed for the proposition pair. It is not clear to us how a prior for a group of people referred to as close patrilineal relatives could be assigned by the court.

We propose that a relevant proposition set is,

Proposition 1. The source of the male DNA is the POI.

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Proposition 2. The source of the male DNA is a random man from the population.

where Proposition 2 is a translation of the generic version "The source of the male DNA is not the POI." to make it operational.

The issue of matching males in the POI's lineage should also be addressed, for example in a comment in the statement. Further (or other types of) investigations can be used to exclude some paternal relatives of the POI if needed.

Using this set, the numerator of the *LR* is indeed 1 for a matching single source Y chromosome haplotype (barring artifacts). The denominator needs to account for the probabilities of obtaining the same profile from anyone who is not the POI. This includes people with all degrees of relatedness from one to a great many steps away [12]. Although estimation is not the focus of this short commentary, we briefly mention that there are suggestions to explicitly model relatedness [11–16]; however, most laboratories still apply frequency estimation methods like the count or augmented count in a database. The count or augmented count will be adequate for common profiles searched in a large database and homogenous populations. It is likely to be inadequate for small databases, inhomogeneous populations, or rare haplotypes which are likely the norm for modern Y-STR profiling systems containing loci with high mutation rates [17–19].

Our suggested proposition pair aligns with the recommendations given by the DNA Commission of the ISFG, and Andersen et al., Andersen and Balding, and Caliebe et al. In addition, although not explicitly given, we believe they align with the sets described by Caliebe and Krawczak, Wolf et al., Krawczak, and the 2001 DNA Commission of the ISFG.

We speculate how the confusion may have arisen. When Y chromosome multiplexes typed only a few loci most patrilineal relatives did indeed share a haplotype. This is less true now both because of the increased number of loci typed but also the inclusion of rapidly mutating markers within forensic multiplexes. However, we speculate that reporting scientists feel the justified need to inform the court of the fact that patrilineal relatives stand an enhanced chance of sharing the haplotype. Unfortunately, this valid statement found its way into the propositions where it now contributes to an invalid set of propositions. We suggest that this valid information should be included in the body of the statement, but not in the propositions themselves. We further speculate that reporting scientists have aligned propositions sets for Y chromosomal evidence with those used for autosomal STR evidence which traditionally considers individuals unrelated to the POI as contributors under the alternate proposition.

4 | CONCLUSION

Unless the circumstances of an individual case suggest differently, there seems to be neither a logical nor a legal basis for changing the prosecution proposition from:

Proposition 1. The source of the male DNA is the POI,

Proposition 1a. The source of the male DNA is the POI or somebody from their lineage.

to:

Excluding the possibility of mutation, the likelihoods of these two propositions would be identical for lineage markers anyway.

We also find no logical nor legal basis for the change in the alternate proposition, from:

Proposition 2. The source of the male DNA is not the POI, to

Proposition 2a. The source of the male DNA is not the POI nor anybody from their lineage.

Pr(*E*|Proposition 2) will be larger and possibly much larger than Pr(*E*|Proposition 2a). The data required to inform Pr(*E*|Proposition 2a) would need to be determined for each case after identifying and subtracting from the database those persons of the same lineage as the POI. This is an almost insurmountable challenge complicated by the impossible task of defining relatedness. Pr(*E*|Proposition 2) may be reasonably informed by a simple sample from the relevant population incorporating relatives if they come into the sample by chance. This is the approach recommended by the UK Regulator where they suggest assigning $\Pr(G_S|P_2,I)$ using $(n+1)/(N+1)$.

It is likely that reference to individuals of the POI's lineage within the propositions is an honest mistake made by laboratories intending to convey the important issue of shared haplotypes among related individuals. We agree this is an important detail to convey but that this point should be made outside of the propositions. This practise is neither logical nor practical, and we do not believe it is the actual statistic being reported.

The count in a representative database gives the sample estimate for Pr(*E*|Proposition 2). If the database is large and represents the generic diversity in the population, then it truly represents a sampling of "random men" and the count will include matching profiles due to relatedness. More likely is that the database does not fulfill these requirements and the count of the haplotype will be 0, thereby still requiring some additional consideration of relatedness to the POI in the population. Explicit modeling of relatedness is a very good way to proceed, and methods applying this approach have been published [11, 16].

Taylor et al. [16] use the effective population size from which the modern population is drawn, Ne, to model the degree of relatedness between any two males. Then, the time to most recent common ancestor (MRCA) can be set by Ne. Ne can be set from any assignment of the match probability for this haplotype.

Andersen and Balding [11] calculate a number of related males with matching haplotypes expected in the population. This could then be incorporated into the LR by stratifying Proposition 2 by

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related males and unrelated males (based on some arbitrary definition of related).

In this work, we limit the discussion of proposition sets to when there is only one POI. Different proposition sets may be required where there are multiple POIs. This could include conditioning on an individual who is reasonably expected to be present in the sample, and compound and complex proposition sets where multiple males could be possible donors.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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